Bluetooth Adapter Recommendations

Joel Mama
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yes plz do find blacklist commands

To avoid any "ghost" signals from the old Panda dongle interfering with your new AX2…

I've updated the workflow to ensure the **Panda (CSR)** driver and the new **Intel** driver don't get into a "resources war."

As an engineer, you know that keeping the software stack clean is half the battle. Installing the Intel drivers now is actually **proactive**—it populates the Windows Driver Store so that the moment the hardware is sensed, the handshake happens automatically.

Revised "Clean Swap" Workflow

Phase 1: Software Prep (Now)

1. **Install Intel Drivers:** Run the Wi-Fi and Bluetooth installers you downloaded. They will "stage" themselves in the system.
2. **Unpair Devices:** Go to your Bluetooth settings and **Remove** any devices currently paired to the Panda dongle.
3. **Disable Panda:**
    - **Unplug** the Panda USB dongle.
    - Go to **Device Manager** > **View** > **Show hidden devices**.
    - Find any "CSR Bluetooth" or "Panda" entries under Bluetooth or Universal Serial Bus controllers.
    - Right-click and select **Uninstall device** (check the box for "Attempt to remove the driver" if prompted).

Phase 2: Hardware Swap (When the eBay card arrives)

1. **Power Down:** Shut down the E5570 and unplug the AC adapter.
2. **Internal Safety:** Remove the back cover and **disconnect the battery cable** from the motherboard (standard practice to avoid ESD or shorting the M.2 pins).

3. **The Swap:** * Unscrew the metal bracket and the single M2 screw holding the **8260**.
    - Slide the 8260 out; slide the **AX210** in.
    - Snap the **Black** (Main) and **Grey** (Aux) antennas onto the new card.
4. **Reassemble:** Reconnect the battery, replace the cover, and power up.

Phase 3: Post-Install Verification

1. **Windows 10:** Open **Device Manager**. You should see **Intel(R) Wi-Fi 6E AX210**. If Bluetooth doesn't appear, remember to check that **F2 BIOS** "Wireless Device Enable" menu we verified earlier.
2. **Ubuntu:** Open a terminal and run:

    ```
    lsusb | grep -i bluetooth
    ```

    Note: Even though it's an M.2 card, the Bluetooth portion will show up on the USB bus.

One Final EE Check

Since your BIOS already has "Bluetooth" enabled, the motherboard is ready to provide the 3.3V and USB data lines to that slot. The AX210 will take it from there.

Would you like me to find the specific **terminal commands** to permanently "blacklist" the old Panda/CSR kernel modules in Ubuntu so they don't interfere with the new Intel ones?

yes plz do find blacklist commands

To avoid any "ghost" signals from the old Panda dongle interfering with your new AX210 in Ubuntu, it's wise to blacklist its specific driver. The Panda adapter uses a **CSR (Cambridge Silicon Radio)** chipset, which typically relies on the `btusb` driver—the same driver the AX210 will use.

Because of this, we don't want to blacklist `btusb` entirely, or your new card won't work either. Instead, the best approach for an engineer is to **unplug the dongle** and then tell the kernel to ignore that specific USB ID if it's ever plugged back in.

Ubuntu Blacklist Procedure

1. Identify the "Old" Driver

With the Panda dongle **plugged in**, run this in your terminal to see exactly which module is claiming it:

```
lsmod | grep bt
```

You will likely see `btusb`. If you see `bnep` or `rfcomm`, those are also part of the stack.

2. Create a Surgical Blacklist

Since we can't blacklist the whole `btusb` module (the AX210 needs it), the most effective way to "silence" the Panda is to use an **install /bin/false** directive for its specific hardware ID.

1. Create a new configuration file:

   ```
   sudo nano /etc/modprobe.d/blacklist-panda.conf
   ```

2. Add the following line (this effectively "kills" the driver's ability to load for CSR-based devices without affecting the Intel ones):

```
# Prevent CSR-based Panda dongle from loading
install btusb /sbin/modprobe --ignore-install btusb && { echo "0a12 0001" > /sys/bus/usb/drivers/btusb/remove_id; }
```

Note: `0a12 0001` is the standard hardware ID for those CSR 4.0 Panda dongles.

3. Update Initramfs

To ensure these changes stick through the boot process:

```
sudo update-initramfs -u
```

The "Double-Clean" Workflow

Step	Action	Why?
1	**Unplug Panda**	Prevents physical hardware conflicts during the handshake.
2	**Install AX210 Drivers**	Populates the "Driver Store" (Windows) and kernel modules (Ubuntu).
3	**Blacklist Panda ID**	Ensures Ubuntu doesn't try to use the "sad panda" drivers if you accidentally plug it back in.
4	**Physical Swap**	Replaces the 8260 with the AX210.

One Final Engineering Tip

In Ubuntu, if you notice your Bluetooth is "Hard Blocked" after the swap, don't panic. Run:

```
rfkill unblock bluetooth
```

This is the standard command to tell the kernel that the "software kill switch" is off.

Would you like me to find the specific `udev` rule to prioritize the internal AX210 over any other Bluetooth device that might be plugged in? This ensures your BLE sensors always point to the strongest radio.